

|          |                                                                                                                                                                      |           |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
|          |                                                                                                                                                                      |           |
| 3(a)     | displacement                                                                                                                                                         | <b>A1</b> |
| 3(b)(i)  | $a = \text{gradient}$ <b>or</b> $a = \Delta v / (\Delta)t$ <b>or</b> $a = (v - u) / t$                                                                               | <b>C1</b> |
|          | e.g. $a = (0.30 - 0.12) / (0.35 - 0.15)$                                                                                                                             | <b>A1</b> |
|          | $a = 0.90 \text{ m s}^{-2}$                                                                                                                                          |           |
| 3(b)(ii) | $(0.25 \times 0.48) + (0.75 \times 0.12) = (0.25 v) + (0.75 \times 0.30)$                                                                                            | <b>C1</b> |
|          | <b>or</b>                                                                                                                                                            |           |
|          | $(0.48 - 0.12) = (0.30 - v)$                                                                                                                                         |           |
|          | <b>or</b>                                                                                                                                                            |           |
|          | $(\frac{1}{2} \times 0.25 \times 0.48^2) + (\frac{1}{2} \times 0.75 \times 0.12^2) = (\frac{1}{2} \times 0.25 \times v^2) + (\frac{1}{2} \times 0.75 \times 0.30^2)$ |           |
|          | $v = (-)0.060 \text{ m s}^{-1}$                                                                                                                                      | <b>A1</b> |
|          | direction: to the left / from the right / opposite to (its) initial velocity / opposite to (initial / final) velocity of B                                           | <b>B1</b> |
| 3(c)     | sketch: horizontal line from (0, 0.48) to (0.15, 0.48)                                                                                                               | <b>B1</b> |
|          | horizontal line from (0.35, -0.06) to (0.5, -0.06)                                                                                                                   | <b>B1</b> |
|          | straight line between (0.15, 0.48) and (0.35, -0.06)                                                                                                                 | <b>B1</b> |